

- ① What is the difference between heuristic and uniformed search methods? Explain the working of Best-First Search and list its advantages.

<u>Feature</u>	<u>uniformed search</u>	<u>Heuristic Search</u>
1. Knowledge used	Has no information about the distance to the goal	use problem specific knowledge (heuristic)
2. Efficiency	Explores state blindly less efficient	More efficient. Narrows the search space
3. Cost	Hightime and memory complexity	Lower complexity in practice
4. Examples	BFS, DFS	A* search, Best-First Search

Best-First Search (BFS)

BFS is an informed search algorithm that uses an evaluation function to decide which node to expand next.

How it works:

1. It maintains two lists: OPEN (nodes to be visited) and CLOSED (already visited)
2. It uses a heuristic function $h(n)$, which estimated the 'closeness' of a node to the goal.
3. At each step, the algorithm selects the node with the lowest heuristic value from the OPEN list.
4. That node is expanded, and its children are added to the OPEN list
5. This process is repeated untill the goal node is found or the OPEN list is empty.

Advantages of Best-First Search

- Faster than uninformed search methods
- Uses heuristic knowledge to guide search
- Reduces unnecessary exploration
- Suitable for large search spaces
- Simple and efficient for many practical problems

- ② Discuss stochastic search methods in AI. compare different approaches such as simulated annealing and genetic algorithms in terms of their working and applications.

Stochastic search methods are algorithms that use randomness (probability) to explore the search space and find optimal or near-optimal solutions.

Unlike deterministic methods, they do not always follow a fixed path and are useful for solving complex problems where the search space is large.

1. Simulated Annealing (SA)

Simulated Annealing mimics the process of heating a metal and then cooling it slowly to reach a low-energy, crystalline state.

— Working mechanism

- The Temperature Factor: The algorithm starts with a temperature (T). At high T , the algorithm is very likely to accept 'bad' moves to explore the landscape
- The cooling Schedule: As the search progresses T gradually decreases
- Convergence: As the 'metal' cools, the algorithm becomes more selective, eventually only accepting moves that improve the solution, settling into a minimum.

- Applications

- VLSI Design
- Travelling Sales person problem
- Job - Shop scheduling

2. Genetic Algorithm (GA)

GA are based on Darwins theory of natural selection. They maintain a population of potential solution rather than just one.

- Working mechanism

- Selection: The fittest individual are chosen to reproduce
- Crossover: portions of two parent solutions are combined to create offspring, hoping to pass on superior traits.
- Mutation: random changes are occasionally introduced to a solution to maintain genetic diversity and prevent the population becoming stagnant.

- Applications

- Bio informatics
- Economics
- Aerospace Engineering